

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · DPP SHEET

Chapter 22: Heights and Distances

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. The angle of elevation of the top of a pole from a point 15 m from its foot is 45° . Find the height of the pole.
2. A ladder 8 m long reaches a point 4 m up a wall. Find the angle it makes with the ground.
3. The shadow of a vertical tower is 20 m long when the sun's altitude is 60° . Find the height of the tower.
4. From the top of a building 30 m high, the angle of depression of a car is 30° . Find the distance of the car from the building.
5. A man 1.6 m tall stands 20 m from a tree and sees its top at an angle of elevation of 45° . Find the height of the tree.
6. The angle of elevation of the top of a tower is 45° . On walking 20 m towards the tower, it becomes 60° . Find the height of the tower.
7. Two men on opposite sides of a tower 60 m high observe its top at angles of elevation 45° and 60° . Find the distance between the men.
8. From the top of a cliff 90 m high, the angles of depression of two boats on the same side, in line with the foot, are 30° and 60° . Find the distance between the boats.
9. From the top of a tower 100 m high, the angles of depression of the top and bottom of a building are 30° and 45° . Find the height of the building.
10. An aeroplane flying at a height of 3000 m passes vertically above another aeroplane. From a point on the ground, their angles of elevation are 60° and 45° . Find the vertical distance between them.

Answer Key

1. 15 m
2. $\sin \theta = \frac{1}{2}$, so 30°
3. $20\sqrt{3} \approx 34.64$ m
4. $30\sqrt{3} \approx 51.96$ m
5. $20 + 1.6 = 21.6$ m
6. $h = \frac{20\sqrt{3}}{\sqrt{3} - 1} = 10(3 + \sqrt{3}) \approx 47.32$ m
7. $60 + 20\sqrt{3} \approx 94.64$ m
8. $90\sqrt{3} - 30\sqrt{3} = 60\sqrt{3} \approx 103.92$ m
9. $d = 100$ m; $100 - \frac{100}{\sqrt{3}} \approx 42.26$ m
10. $d = \frac{3000}{\sqrt{3}} = 1000\sqrt{3}$; lower plane at 1732 m;
distance ≈ 1268 m